MILITARY SPECIFICATION

COMPRESSORS, HERMETIC: FOR USE WITH REFRIGERANT 12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hermetic compressors in a nominal horsepower range of 1/3 to 7-1/2 inclusive.

1.2 Classification. Hermetic compressors shall be of the following sizes, as specified (see 6.2):

Sizes	Nominal compressor capacities for condensing temperature + 135 F and the following saturated refrigerant suction temperatures at:		
	-10 F	+25 F	+50 F
	Btu/hr	Btu/hr	Btu/hr
H-1	900	2,500	------
H-2	1,200	3,400	------
H-3	1,800	4,800	------
H-4	2,400	6,500	10,700
H-5	4,600	13,700	23,000
H-6	5,900	16,800	30,500
H-7	9,000	29,200	50,700
H-8	12,400	40,200	73,000

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2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
PPP-B-601	-	Boxes, Wood, Cleated-Plywood

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-773	-	Compressors, Hermetic: For Use with Refrigerant 12
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MS 17248	-	Valves, Compressor Service, Refrigerant 12

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR

CONDITIONING ENGINEERS, INCORPORATED (ASHRAE)

Standard No. 23
Methods of Testing for Rating Refrigerant Compressors

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, N.Y. 10017.)

UNDERWRITERS LABORATORIES INC, (UL)

UL Standard No. 303 - Refrigeration and Air Conditioning Condensing and Compressor Units
UL Standard No. 984 - Sealed (Hermetic type) Motor-Compressors

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The compressors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

* 3.2 Standard product. The compressors delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with this specification. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

* 3.3 Codes and standards. The compressors shall comply with the applicable requirements of UL No. 303 and UL No. 984.

* 3.3.1 Compliance. Prior to approval of the first shipment of compressors the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the compressors he proposes to furnish under this specification meet the applicable requirements of UL No. 303 and UL No. 984.

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* 3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a Recognized Component Mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the compressors have been tested and conform to UL No. 303 and UL No. 984.

* 3.4 Materials. Materials and components shall be as specified herein and in MIL-STD-773. Materials not definitely specified shall be of the quality normally used by the manufacturer for hermetic compressors, provided the completed item complies with all provisions of this specification.

3.5 Design and construction. The compressors shall be designed and constructed in accordance with the requirements specified herein and in MIL-STD-773.

3.5.1 Electrical characteristics. Unless otherwise specified (see 6.2), electrical characteristics for each size hermetic compressor shall be as specified in MIL-STD-773.

3.5.1.1 Electromagnetic compatibility. Compressors shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.6.1.7).

3.5.2 Capacities.

3.5.2.1 Refrigeration capacities. Compressors shall have the capacity indicated in 1.2, plus or minus 15 percent (see 4.6.1.2) and shall be capable of repeating capacity test results within plus or minus 3 percent (see 4.6.1.3).

3.5.2.2 Pumping capacity. Compressors shall have a pumping capacity not more than 5 percent below the standard determined in 4.6.2.2.2 when tested as specified in 4.6.2.3. The power input shall be not greater than 10 percent above the standard determined in 4.6.2.2.2 when tested as specified in 4.6.2.3.

* 3.5.3 Refrigerant. The compressors shall be designed for use with refrigerant conforming to type 12 of BB-F-1421.

3.5.4 Operation.

3.5.4.1 High and low temperature. Compressors shall operate without breakage, leaks, knocking, rocking, or objectionable noise in ambient temperatures from minus 25F to plus 125F when tested as specified in 4.6.1.5.

3.5.4.2 Endurance. There shall be no leaks, breakage, or operating noise due to wear of mechanical equipment and there shall be no tripout of any control device when tested as specified in 4.6.1.4.

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3.5.4.3 Low-voltage start. Compressors shall be capable of starting when electric characteristics are such that the frequency is 100 percent of rated value, line starting voltage is 85 percent of rated voltage, and refrigerating conditions are 4.5 psig (minus 10 F saturated refrigerant vapor suction temperature) and 197.5 psig (135 F saturated) condensing pressure when tested as specified in 4.6.1.6.

3.5.5 Storage. Compressors shall be capable of withstanding storage temperatures ranging from minus 65 F for a period of at least 3 days duration, to 155 F for periods as long as 4 hours per day. There shall be no deterioration, leakage, entrance of moisture, or other impairment as a result of storage conditions when tested as specified in 4.6.1.1.

3.5.6 Service valves. Each compressor shall be supplied with discharge and suction service valves conforming to MS 17248. The valves shall be bolted to a mounting plate as shown on figure 6 of MIL-STD-773.

3.5.7 Base and mounting plates. The compressor shall be bolted on an aluminum base shown on figure 6 of MIL-STD-773. The connection mounting plate shall be attached to the base plate.

3.6 Marking for identification. In addition to the markings required by UL Standard No. 303, compressors shall be marked in accordance with MIL-STD-130.

3.7 Finish. Metal surfaces shall be thoroughly cleaned, conditioned, and primed in accordance with standard commercial practice. After the primer has thoroughly dried, surfaces shall be given a coat of semi-gloss enamel conforming to TT-E-529. Unless otherwise specified (see 6.2), the color shall be olive drab conforming to number 24087 of FED-STD-595. The enamel shall be free of streaks, drops, pinholes, checks, or cracks.

3.8 Workmanship. Soldering shall be complete, clean, adherent, and shall not have any pinholes. Bolts, nuts, screws, studs, and other types of threaded fasteners when used, shall not be broken, stripped, fractured, or loose and shall have lock washers or shall be a self-locking type.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections.

a. Qualification inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 Qualification inspection. Qualification inspection shall consist of all the tests specified in 4.6.1. The test specified in 4.6.1.4 shall be performed last. No order is specified for the other tests. One compressor of each size submitted under this specification shall be required to pass the qualification tests. Failure to pass any test shall be cause for rejection of qualification.

4.4 Quality conformance inspection. Sampling for this inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Examination shall be made to insure that surface preparation and priming is accomplished as required under 3.7. When nonconformance is noted, correction shall be made to the affected end items and to the operation.

4.4.3 End item inspection. The inspection lot shall consist of all compressors of one size offered for inspection at one time. The sample unit shall be one compressor.

4.4.3.1 Visual examination. Examination of the compressor shall be in accordance with the classification of defects set forth in table I. The inspection level shall be II with an acceptable quality level (AOL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not specified color		X
	Streaks, drops, pinholes, checks, or cracks		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction, and workmanship	Part missing or malformed (unless otherwise classified herein)	X	
	Part not connected or joined as specified	X	
	Threaded fasteners broken, stripped, fractured, or loose		X
	Lockwashers missing where required		X
Soldering	Soldering missing, incomplete, or not adherent	X	
	Soldering not clean (flux or flux residue not removed) or pinholes in solder		X
Marking	Identification marking missing, not legible, not specified type, not affixed in the prescribed manner		X

4.4.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that does not meet specified requirements shall be classified a defect. The inspection level shall be S-2 with an AOL of 4.0, expressed in terms of defects per hundred units.

4.4.3.3 End item testing. Each compressor shall be tested as specified in 4.6.2.3. Failure of the compressor to pass this test shall be cause for rejection of the compressor.

4.4.4 Examination of packaging. An examination shall be made to determine that packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AOL of 4.0, expressed in terms of defects per hundred units.

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TABLE II. Packaging defects

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing, damaged or otherwise defective

4.4.4.1 Inspection of shipping containers. When shipping containers are required to be in accordance with PPP-B-601, examination for defects in closure and strapping shall be in accordance with appendix of PPP-B-601. Examination for defects in cushioning, blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186.

4.5 Code and standards compliance. Proof of compliance with the UL requirements of 3.3 shall be available to the Government representative.

4.6 Tests.

4.6.1 Qualification testing. Qualification tests shall consist of tests specified in 4.6.1.1 through 4.6.1.7.

4.6.1.1 Storage test. To determine compliance with the storage requirements of 3.5.5, the compressor, complete with oil and refrigerant holding charge, shall be stored in an ambient temperature of minus 65 F for a period of at least 3 days duration, and then in an ambient temperature of 155 F for 4 hours. Compressor may be stored at room temperature for 24 hours between the two periods of extreme temperature. Any leakage deterioration, or entrance of moisture shall constitute failure of this test.

4.6.1.2 Capacity tests. Conditions for this test shall be as follows:

(a) Rated voltage specified, plus or minus 2 volts
(b) Frequency: 60 Hz (Hertz), plus or minus 1/2 Hz
(c) Ambient temperature: 110 F
(d) Condensing temperature: 135 F
(e) Suction vapor temperature (runs shall be made at the following temperatures):

Size	Saturated	Actual
All	- 20 F	40 F
All	- 10 F	50 F
All	25 F	65 F
H-4 to H-8	50 F	65 F

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The test shall be conducted in accordance with ASHRAE Standard No. 23 except as specified herein. Capacities obtained during the test shall be corrected to 0 F subcooling and shall be checked for compliance with 3.5.2.1. Inability of the unit to meet capacity requirements of 3.5.2.1 shall constitute failure of this test.

4.6.1.3 Verification capacity test. A verification capacity test shall be run in accordance with the procedures set forth in 4.6.1.2 using minus 10 F saturated suction vapor temperature conditions only. Inability of unit to repeat capacity within plus or minus 3 percent shall constitute failure of this test.

4.6.1.4 Life test. Test conditions and operating time shown in table III shall be used for this test to determine conformance with the mechanical equipment, endurance, and tripout requirements of 3.5.4.2. The test shall be conducted in a closed system and after stable operating conditions are reached. The following data shall be recorded hourly:

(a) Ambient temperature

(b) Condensing temperature

(c) Saturated refrigerant suction vapor temperature

(d) Actual suction vapor temperature

(e) Voltage

(f) Frequency

(g) Amperes

(h) Watts input

(i) Power factor

After running the test, the compressor shall be completely disassembled and all the parts examined for excessive wear, damage, or distortion. Inability to meet requirements shall constitute failure of this test.

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TABLE III. Life test conditions

Test designation	Duration hours	Operation C-Continuous 1/ I-Intermittent 2/	Voltage 3/	Frequency (Hertz) ± 1/2	Ambient temperature	Condensing temperature	Saturated suction temperature	Actual suction temp.
a	100	C or I	Std	60	100-110	135	-10	50
b 4/	100	C or I	Std	60	100-110	135	50	65
c 4/	8	C	90% Std	60	100-110	135	50	65
d 4/	8	C	110% Std	60	100-110	135	50	65
e 4/	8	C	Std	60	100-110	135	50	65
f 4/	8	C	Std	60	100-110	135	50	65
g	8	C	Std	60	110 Min	135	-10	50
h 4/	8	C	5/	60	110 Min	135	50	65

1/ Continuous operation is operation at stable conditions without shut-down.

2/ Intermittent operation is made up of continuous operation at stable conditions for periods of not less than seven hours.

3/ Standard voltage: Rated voltage specified plus or minus 2 volts.

4/ Size H-1, H-2, and H-3 shall be operated at 25 F saturated suction temperature in lieu of temperature specified in table III for tests b, c, d, e, f, and h.

5/ Vary voltage between 90 and 110 percent of standard and observe current. Use voltage which gives maximum current.

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4.6.1.5 Temperature tests.

4.6.1.5.1 High temperature. The compressor shall be operated for a period of 4 hours and the test results checked for conformance with 3.5.4.1. Conditions for the test shall be as follows:

a. Voltage: Rated voltage specified, plus or minus 2 volts

b. Frequency: 60 Hz (Hertz), plus or minus 1/2 Hz

c. Ambient temperature: 125 F

d. Condensing temperature: 155 F

e. Saturated refrigerant suction vapor temperature: 25 F for sizes H-1 through H-3. 50 F for sizes H-4 through H-8

f. Actual suction vapor temperature: 65 F

Inability of the compressor to operate without breakage, leakage, or noise caused by wear shall constitute failure of this test.

4.6.1.5.2 Low temperature. The compressor shall be operated under the same test procedure and conditions of 4.6.1.5.1 except that the ambient temperature shall be minus 25 F and the condensing temperature 105 F. Any breakage, leaks, motor winding temperature above 302 F, resistance to ground less than 100 ohms, or tripouts, shall constitute failure of this test.

4.6.1.6 Adverse starting test. The compressor shall be operated under the conditions of 4.5 psig (minus 10 F. saturated refrigerant suction vapor temperature) and 197.5 psig (135 F saturated) until the condensing pressure is stable. The frequency shall be adjusted to 60 Hz and the line voltage adjusted so that the starting voltage caused by the inrush current is 85 percent of rated value. After these conditions have been attained, the compressor shall be stopped and then started within one minute and then allowed to run for 10 minutes. Five such starts shall be made to determine conformance with the starting requirements of 3.5.4.3. Failure of the motor to start shall constitute failure of this test.

* 4.6.1.7 Electromagnetic compatibility test. The initial unit of production shall be tested by the supplier in accordance with test methods CE03, CE05, and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish to the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Test Report and the requirements meet MIL-STD-461.

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4.6.2 Capacity tests.

4.6.2.1 Capacity check test. A refrigeration capacity check test shall be performed using either the primary or confirming capacity test method of ASHRAE Standard 23 to determine compliance with 3.5.2.1 except a repeat test is not required. The refrigeration capacity in B.t.u./hour shall be determined at conditions of minus 10 F and plus 25 F saturated refrigerant suction vapor temperatures and 135 F condensing temperature. Failure to meet capacity in 1.2 plus or minus 15 percent shall constitute failure of this test.

4.6.2.2 Pumping capacity standard test. Upon successful completion of the test specified in 4.6.2.2.1, the compressor passing the test shall be used to determine pumping standards as follows: The compressor shall be set up on a calibrated flow test device normally used for compressor capacity flow testing. The compressor shall be run three times at design voltage and frequency conditions. The flow capacity and watt input for each run shall be recorded and averaged. This average shall then be used as a standard to determine acceptability of the compressors tested in 4.6.2.3.

4.6.2.3 Pumping capacity test. Each compressor shall be tested in accordance with the test procedure specified in 4.6.2.2.2, except the compressor shall be run once. The test results shall be checked to determine compliance with 3.5.2.2. Pump capacity more than 5 percent below or power input more than 10 percent above the standard determined in 4.6.2.2.2 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each compressor shall be packed in a shipping container conforming to overseas type, style A or B, type 3 load of PPP-B-601. Cushioning, blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186. Each shipping container exceeding 250 pounds gross weight shall be provided with skids in accordance with PPP-B-601. Closure and strapping shall be in accordance with appendix of PPP-B-601.

5.1.2 Level B. Each compressor shall be packed as specified in 5.1.1, except that the shipping container shall conform to domestic type, style A or B, type 3 load of PPP-B-601.

* 5.1.3 Level C. The compressor shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies, provided the quantity packed per shipping container is the same as that normally used by the supplier for retail distribution.

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5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Military standard hermetic compressors are intended for use in combination with other Military standard refrigeration components in refrigeration and air-conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size compressor required (see 1.2).
(c) Electrical characteristics, if other than specified (see 3.5.1).
(d) Color of finish, if other than specified (see 3.7).
(e) Selection of applicable level of packing (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Oualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be elegible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the U.S. Army Natick Research and Development Command, Natick, MA 01760, and information pertaining to qualification of products may be obtained from that activity.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
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Review activity:

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User activity:

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